

Qns	Answer	Marks
4(a)(i)	current in solenoid produces magnetic field with field lines parallel to axis of solenoid	1
	radii of copper disc cuts magnetic flux as it rotates,	1
	(by Faraday's law) rate of change of magnetic flux linkage results in emf	1
4(a)(ii)	consider a radial strip of copper: by faraday's law: $ E = \frac{d}{dt}(N\phi) = \frac{d}{dt}(NBA)$ $= (1)(B) \frac{d}{dt} \left(\frac{1}{2} R^2 \theta \right)$ $= \frac{1}{2} BR^2 \omega$ $= \frac{1}{2} BR^2 (2\pi f)$ $= BAf$	1 1
4(b)(i)	at null deflection, p.d. across resistor = induced e.m.f. $V_R = E$ $IR = BAf$ $= (\mu_0 n I) Af$ same current in solenoid and resistor BUT NOT COPPER (disc/axle) $R = \mu_0 n Af$	1
4(b)(ii)	no electrical quantities needed so not dependent on accuracy of any voltmeter, ammeter or ohmmeter used	1
4(c)	$BAf = (\mu_0 n I) Af \quad \text{--- (1)}$ $R = \mu_0 n Af \quad \text{--- (2)}$ take $\frac{(1)}{(2)}$: $\frac{BAf}{R} = \frac{(\cancel{\mu_0 n I}) \cancel{Af}}{\cancel{\mu_0 n Af}}$ $B = \frac{IR}{Af} = \frac{(1.0 \times 10^{-3})(10)}{(\pi(0.20^2))(5.0)}$ $= 0.0159 \text{ T}$	1

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Qns	Answer	Marks
5(a)	1	